Joshua Buttrick

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Education

Worcester Polytechnic Institute

- Mechanical Engineering Major
- Presidential Scholarship Recipient

Work Experience

Valco Instruments

- Engineering Internship - Summer 2026
    - Programmed and ran CNC machines mostly involving subspindle turning and 5 axis milling

Breakthrough Miami

- Teaching Fellow - Summer 2024 - Summer 2025
    - 6th grade science and robotics teacher
    - 9th grade science and computer aided design teacher

Andymark Robotics Curriculum Developer - 2023-2024

- Developed an introductory middle school curriculum for the Adymark Robits FTC kit

Projects

- Interactive Qualifying Project: Identifying and Overcoming Language Barriers in Nordural Aluminum's Manufacturing Workplace in Iceland
- Sustainable Skylines: Team research project on the effect of climate change on major U.S. Northeast cities
- Design of Machine Elements: Team project to design and perform static and dynamic fatigue analysis of a wheelchair exercise machine

Organizations and Teams

Formula SAE

- Wheel assembly design and manufacturing lead

Academic Robotics

- First Robotics, Team 190: design and manufacturing mentor mentor, responsible for teaching high school students CAD, CAM, and how to operate manual and CNC machines

WPI Combat Robotics Team

- Design, build, and compete with 1, 3, and 30 pound combat robots

Technical Skills

- CAD: Advanced skills in Solidworks, Fusion 360, and Onshape
- Machining: Extensive Knowledge in Fusion 360 CAM and operating CNC Mills and Lathes. Proficient in manual milling and lathing.
- Finite Element Analysis: Skills in Solidworks and Ansys
- 3D Printing: Extensive knowledge in designing for FDM and SLS manufacturing
- Other skills: Google Sheets/Excel, Soldering

Relevant Courses

Design of Machine Elements, Kinematics of Mechanisms, The Finite Element Method, Advanced Dynamics, Mechanical Applications in Robotics, Material Sciences, Engineering Experimentation